Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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| Title of Proposed Technology | First-in-Class Antibody and Immunocytokine Platform for Tumor Microenvironment Reprogramming | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | Bio·Medical | | Bio·Medical | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | Medical Bio | Therapeutic Antibodies | Medical Bio | Protein Therapeutics |
| Concept of Proposed Technology | The proposed technology is a **first-in-class antibody-based immunotherapy platform** targeting a **novel myeloid-associated immune checkpoint** with no approved therapeutic precedents. The platform is designed to modulate the **tumor microenvironment (TME)** by directly engaging myeloid immune cells, particularly monocytes and macrophages, to induce **pro-inflammatory (heretofore referred to as "M1") reprograming**, enhance immune cell recruitment, and promote effective antitumor immune responses.<br>The core antibody component drives differentiation of monocytes toward an M1 macrophage phenotype, activates T cells to induce cytokine secretion, and promotes tumor cell apoptosis. To further amplify and spatially localize immune activation within tumors, the antibody is engineered into **fusion-based bispecific constructs** through linker-mediated conjugation with **cytokine or chemokine molecules**. | | | |

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| | Two complementary immunocytokine strategies are pursued:<br>1. **Chemokine–Antibody Fusion Strategy**<br>Incorporation of a chemokine component promotes recruitment and infiltration of immune cells, including T cells and antigen-presenting cells, into the tumor microenvironment. When combined with antibody-mediated myeloid activation, this approach is designed to generate synergistic antitumor activity through coordinated immune cell trafficking and activation at the tumor site.<br>2. **Cytokine–Antibody Fusion Strategy**<br>Fusion with a cytokine component enhances $CD8^+$ T-cell activation, survival, and expansion, strengthening tumor-specific cytotoxic immune responses. In combination with myeloid activation and T-cell stimulation, this strategy supports sustained immune functionality within the TME, enabling robust and durable antitumor efficacy.<br>By selectively delivering cytokine or chemokine signals via myeloid cell–associated targets, the platform enables **localized immune activation** within tumors while minimizing systemic cytokine exposure and associated toxicities. Collectively, this technology represents a **novel myeloid-driven immunocytokine platform** with strong potential as a next-generation anticancer therapeutic approach. |
| **Rationale for Joint R&D Project** | Development of a **first-in-class myeloid-targeting immunocytokine platform**, including bispecific fusion antibodies, requires tight integration of **target biology validation, functional immunology, advanced antibody engineering, and translational oncology expertise**, which cannot be achieved by a single organization alone.<br>The bispecific fusion antibody strategy introduces additional scientific and technical complexity beyond conventional monoclonal antibody development, including linker optimization, dual-functionality validation, and immune synergy assessment.<br>Joint R&D collaboration enables:<br>• Integration of human tumor immunology insights with advanced antibody and fusion protein engineering<br>• Access to clinically relevant human samples, mouse models, and specialized immune functional assays<br>• High-parameter profiling expertise to phenotype macrophages with appropriate marker sets<br>• Accelerated early proof-of-concept generation at research and preclinical stages<br>This collaborative framework significantly reduces technical and translational risks while enhancing feasibility of first-in-class immunocytokine innovation. |
| **Key Words** | 1. First-in-class myeloid-targeted immunotherapy<br>2. Cytokine/Chemokine-fusion antibody<br>3. Myeloid immune modulation<br>4. M1 macrophage polarization and activation<br>5. Tumor microenvironment reprogramming |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level (2) | End of the Project | Level (4) |
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| **R&D Objectives** | The objective of this R&D project is to establish **early-stage proof-of-concept** for a novel myeloid-targeting bispecific immunocytokine platform by:<br>• Validating the immunological relevance and mechanism of action of a novel myeloid-associated target<br>• Designing and evaluating bispecific antibody constructs incorporating cytokine or chemokine molecules<br>• Demonstrating macrophage activation (M1 polarization), immune cell recruitment, and enhanced antitumor immune responses<br>• Establishing a mechanistic and translational rationale to support subsequent non-clinical development | | | |
| **R&D Target Specifications** | • ≥2-fold increase in M1 macrophage–associated markers relative to control or monotherapy<br>• Statistically significant enhancement of immune cell activation and recruitment in in vitro and ex vivo assays<br>• Demonstration of target-dependent immune activation with minimal off-target signaling<br>• Confirmation that bispecific antibody constructs exhibit superior efficacy compared to single-agent approaches | | | |
| **R&D Plan** | **Phase 1: Target Biology and Mechanism Definition**<br>• Characterize expression and functional roles of the myeloid-associated target in monocytes and macrophages<br>• Evaluate modulation of immunosuppressive TME conditions through macrophage polarization and immune signaling analyses<br>• Establish mechanistic hypotheses supporting cytokine and chemokine fusion strategies<br><br>**Phase 2: Bispecific Antibody and Immunocytokine Design**<br>• Engineer and optimize bispecific fusion antibodies using linker-based molecular designs<br>• Evaluate dual-binding functionality, structural integrity, and molecular stability<br>• Select lead constructs with favorable biophysical and functional profiles<br><br>**Phase 3: Functional In Vitro and Exploratory In Vivo Evaluation**<br>• Conduct in vitro assays to assess macrophage polarization, T-cell activation, and tumor cell killing<br>• Evaluate immune modulation under TME-mimicking conditions<br>• Perform exploratory in vivo studies assessing antitumor efficacy, immune infiltration, PK profiles, and preliminary safety and tolerability | | | |

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | • Validate target relevance and immune modulation mechanisms using human-derived samples<br>• Design and execute in vitro and ex vivo functional immune assays<br>• Assess macrophage polarization, immune activation, and cell recruitment using high-parameter phenotyping, preferably with spatial information<br>• Validate immunomodulation and macrophage reprogramming in vivo using mouse models (applicable if designed assets are cross-reactive)<br>• Advise on clinical feasibility, indication relevance, and translational biomarker strategies |
| **Expected R&R of Korean SME Partner** | • Lead design, engineering, and optimization of antibody and immunocytokine fusion constructs<br>• Conduct early proof-of-concept studies demonstrating target-dependent immune modulation<br>• Execute exploratory in vivo and/or ex vivo efficacy studies<br>• Integrate PoC, efficacy, PK, and safety data to support lead selection and development decisions |

## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | **Expected Outcomes**<br>• Establishment of early proof-of-concept for a first-in-class myeloid-targeting immunocytokine platform<br>• Identification of development-ready lead bispecific antibody candidates<br>• Generation of coherent pharmacological, efficacy, and preliminary safety data packages<br><br>**Expected Impact**<br>• Validation of a modular and expandable myeloid-driven immunotherapy platform<br>• Strengthening of Korean SME competitiveness in advanced antibody and immuno-oncology technologies<br>• Creation of high-quality data and intellectual property supporting future partnerships and licensing opportunities<br>• Contribution to national Advanced Bio innovation and development of novel therapeutic options for patients with unmet medical needs |

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## 6. References

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(2) Lee S, Lim CK, Kim J, Kim J, Jin HK, Bae JS & Jeon JW. Engagement of CD300c by a novel monoclonal antibody ameliorates behavioral deficits in a 5xFAD mouse model of Alzheimer's disease. Biomedicines 13, 1169 (2025).

(3) S. Kim *et al.* The combination of CD300c antibody with PD-1 blockade suppresses tumor growth and metastasis by remodeling the tumor microenvironment in triple-negative breast cancer. *Int. J. Mol. Sci.* **26**, 5045 (2025).

(4) Kim S., Han I.-H., Lee S., Park S., Jeon J.-W. & Bae H. The CD300c antibody CL7 suppresses tumor growth by regulating the tumor microenvironment in non-small cell lung carcinoma. *Front. Oncol.* **15**, 1698857 (2025).

(5) A novel antibody against CD300c ameliorates cognitive deficits and reduces pathology in the late-stage of APP/PS1 mouse model